



Figure 5: (a)-(b) Attack performance with different $\mathcal{L}_{inv}$ and top- n selection strategy, respectively. (c) Attack Acc and FID with constraints of various strengths in Eq. (3). (d) Attack Acc and FID with various numbers of augmentations in Eq. (4). (c) and (d) use FaceScrub as the public dataset.

It should be emphasized that in the case of using FaceScrub as the public dataset, both GMI and KED-MI only obtain an almost failed attack performance. However, our method still maintains a high attack success rate, on average 50% higher than KED-MI on Attack Acc. As for the FID, our method maintains a low value in all cases, with an average reduction of about 3 times relative to KED-MI. The lower KNN Dist also shows that our method can reconstruct more accurate private images.

The results of FFHQ as public data are generally better than those of FaceScrub. The possible reason is that FaceScrub consists of face images crawled from the Internet, which is more different from the distribution of CelebA. Furthermore, we find that the degree of privacy leakage under MI attacks varies between model architectures. In our experiments, ResNet-152 is more likely to leak information in private datasets compared to VGG16 and Face.evoLve. This phenomenon deserves further study in future work.

Generality of the GAN. The adversary may simultaneously perform MI attacks against multiple available target models to verify the correctness of the reconstructed image for the target identity. However, it will be very time-consuming to train a GAN separately for a specific target model each time. Therefore, we explore a new scenario where the architecture of the classifier used to help train the GAN is different from that of the target model.

Table 3 compares the results of our method with those of the baselines. Our method yields clearly superior results for all three models under various evaluation metrics. When using VGG16 to provide prior information to train a generator in stage-1, and using it to attack ResNet-152 in stage-2, our method achieves 81% Attack Acc, while KED-MI only achieves 12%. In addition, the FID of KED-MI is approximately 3 times higher on average than ours.

The experimental results of this scenario illustrate that regardless of the architecture of the models, our method is able to extract general knowledge to help recover sensitive information. Specifically, the generality of our cGAN can greatly improve the efficiency of attacks when there are multiple target models to be attacked.

Ablation study. We further investigate the effects of the various components and hyperparameters in PLG-MI and

conduct attacks against VGG16 trained on CelebA using FaceScrub or FFHQ as the public dataset. As shown in Fig. 5 (a), we compare the impact of different $\mathcal{L}_{inv}$ used in MI attacks. Using the max-margin loss brings a significant improvement on Attack Acc, especially when the distributional shift is larger (i.e., the public dataset is FaceScrub). Fig. 5 (b) presents the results of using different n in the top- n selection strategy, its value has no significant impact on the attack performance, indicating that the strategy is not sensitive to n . Fig. 5 (c) shows the attack performance and visual quality when imposing explicit constraints of various strengths on the generator and $\alpha = 0.2$ is optimal among them. Fig. 5 (d) shows that the data augmentation module in stage-2 has a great influence on improving the Attack Acc and the visual quality of reconstructed images. However, when the number of augmentations is greater than 2, the improvement in attack performance is small, but the attack efficiency will be reduced. Thus we finally force the reconstructed images to maintain identity consistency after 2 random augmentations.

5 Conclusion

In this paper, we propose a novel MI attack method, namely PLG-MI attack. We introduce the conditional GAN and use pseudo-labels provided by the proposed top- n selection strategy to guide the training process. In this way, the search space in the stage of image reconstruction can be limited to the subspace of the target class, avoiding the interference of other irrelevant features. Moreover, we propose to use max-margin loss to overcome the problem of gradient vanishing. Experiments show that our method achieves the state-of-the-art attack performance on different scenarios with various model architectures. For future work, the black-box MI attack is still in its infancy, and the idea of our method can be transferred to the black-box scenario to further improve its performance.

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